

Multi-objective optimization of vehicle body B-pillar lower joints based on crashworthiness analysis

Xuejing Du¹ , Zhenzhen Chen¹, Jiali Song² and Zhanyu Wang¹

Abstract

The joints in an automobile's body structure are crucial in bearing loads and transmitting stresses, thereby significantly affecting the body's rigidity. To effectively improve body rigidity and crashworthiness, this study employed a sensitivity analysis to identify the critical joints among the nine joints of a specific sport utility vehicle (SUV) body. Following regulatory requirements, collision simulations were performed, revealing that the joint below the B-pillar exhibited the most significant deformation. Thus, using the material and thickness of the B-pillar's lower joint as design variables, experimental samples were generated by the design of experiment (DOE). A multi-objective optimization for the B-pillar's lower joint model was conducted using the response surface method and the simulated annealing algorithm to determine the final optimized solution. The optimization results showed a 9.31% increase in body bending stiffness, an 11.37% increase in torsional stiffness, and reduced intrusion at various points on the B-pillar, effectively enhancing the body's rigidity and crashworthiness.

Keywords

Body joint structure, body crashworthiness, sensitivity analysis, multi-objective optimization

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Introduction

Being a critical factor in determining the overall rigidity of a vehicle, the rigidity of the body is influenced significantly by the joint structures within the body, accounting for 60% of the body's rigidity. During the conceptual design phase of automobiles, there is substantial flexibility in designing body joints, and a well-designed joints can significantly enhance the body's rigidity, as well as its noise, vibration and harshness (NVH) and strength performance.^{1,2}

Many efforts have been made to optimize the structural design of the B-pillar and various joints in the vehicle body. Izanloo and Khalkhali³ aimed to achieve lightweight and high rigidity for the vehicle body and developed a simplified model of the body-in-prime (BIP), along with an optimization procedure for the B-pillar's design. Yin et al.⁴ established a finite element

model for seven connecting joints in a white body and performed rigidity calculations for the lower joint of the B-pillar. They discussed the influence of sheet metal thickness and rigidity on white body bending, torsion, and first-order bending and torsion frequencies through the use of regional sensitivity analysis and strain energy calculations. Hou et al.⁵ proposed a multi-objective design optimization method for optimizing geometry and lay-ups for carbon fiber reinforced plastics (CFRP)

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T-joints in automobiles aimed at achieving car body lightweight design. Jia et al.⁶ improved joint structures through joint fatigue analysis and real-road tests. Kiani et al.⁷ studied the impact of joint stiffness on the vehicle's vibration response and concluded that increasing joint stiffness was beneficial for improving the vehicle's vibration response performance. Gaeta et al.⁸ employed the contribution analysis method to identify the joints with the most significant impact on the body's free modes. They subsequently optimized these joints using shape optimization techniques, leading to enhancements in the body's free modes. Tunç and Endur⁹ devised a novel approach to quantify the sensitivity of crucial joints in automobiles to critical performance indicators. Subsequently, they utilized these sensitivity results to enhance body performance indicators, illustrating their methodology on a finite element model of the 2010 Toyota Yaris.

Simultaneously, researchers have made numerous attempts to optimize the vehicle structure to enhance its crashworthiness. Xiong et al.¹⁰ proposed a conjoint method that integrates the Hybrid Contribution Analysis (HCA) method, the Artificial Neural Network (ANN) meta-model, the modified Non-dominated Sorting Genetic Algorithm II (MNSGAI), and the Ideal Point Method (IPM). This method is employed for multi-objective optimization of lightweight and crashworthiness in the side structure of automobile bodies. Pyrz et al.¹¹ utilized the macro-element method to model crushing parameters of S-frame, optimizing the crashworthiness of thin-walled frames. Kongwat et al.¹² proposed a method to design an efficient lightweight frame layout and member sections of a bus superstructure, meeting requirements for bending stiffness, torsion stiffness, and rollover safety in accordance with ECE-R66 standards, utilizing optimization techniques. Zhang et al.¹³ combined particle swarm optimization with artificial immune algorithms to address the multi-objective optimization problem of reducing mass while enhancing rollover crashworthiness in trucks. Wang et al.¹⁴ proposed a hybrid optimization method that integrates contribution analysis, Entropy Weight Technique for Ordering Preferences by Similarity to Ideal Solution (E-TOPSIS), design of experiments (DOEs), and grey relational analysis (GRA). This method aims to optimize the design of front-end safety parts of automobile bodies, enhancing both crashworthiness and lightweight characteristics. Zou et al.¹⁵ proposed the negative Poisson's ratio B-pillar optimized by non-dominated sorting genetic algorithm II (NSGA-II), achieving superior comprehensive crashworthiness and effectively enhancing the side crashworthiness of automobiles.

In summary, significant achievements have been made in the research on body joints and crashworthiness, specifically focusing on the B-pillar and other

joints. However, further investigation is necessary to understand the sensitivity analysis of different joint structures regarding the crashworthiness of the vehicle body and their interrelationship. Consequently, this study utilized the finite element model of the vehicle body to identify critical joints affecting the body's bending and torsional stiffness via sensitivity analysis. It also conducted collision simulations to determine the joint experiencing the most severe deformation in a collision. A multi-objective optimization was performed on this joint to mitigate intrusion and deformation during collisions, thereby enhancing the body's stiffness and crashworthiness.

The remainder of this paper is organized as follows: Section "Key body joints analysis" identifies critical joints in the vehicle body via sensitivity analysis and collision simulations. Section "Multi-objective optimization design of joints" discusses the multi-objective optimization of these critical joints. The discussion and conclusion are presented in Sections "Discussion" and "Conclusion", respectively.

Key body joints analysis

Body joints

Joints in the body refer to the transitional regions connecting the body's crossbeams, longitudinal beams, and pillars, typically with two or three branches. In the structure of an SUV body, there are nine pairs of critical joint structures, namely the body's A-pillar, B-pillar, C-pillar, D-pillar, and the connecting areas between the body's crossbeams and longitudinal beams. These include A1 (upper A-pillar joint), A2 (middle A-pillar joint), A3 (lower A-pillar joint), B1 (upper B-pillar joint), B2 (lower B-pillar joint), C1 (upper C-pillar joint), C2 (lower C-pillar joint), D1 (upper D-pillar joint), and D2 (lower D-pillar joint). For instance, the locations of these joints in a specific SUV body are depicted in Figure 1.

Sensitivity analysis of joints based on SUV body rigidity

The study selected a typical mid-size SUV measuring approximately 4.9 m in length, 1.85 m in width, and around 1.75 m in height. The constructed SUV body model was imported into the ANSA software for analysis. Meshing of the model was completed using quadrilateral and triangular elements with an 8 mm grid size, resulting in a finished finite element model of the body, as depicted in Figure 2. Notably, the primary division of the B-pillar predominantly utilized CQUAD4 elements, comprising a total of 13,768 elements.

In this study, a variety of materials were employed in the construction of the SUV model. The material properties of the B-pillar (DC03) are as follows: the

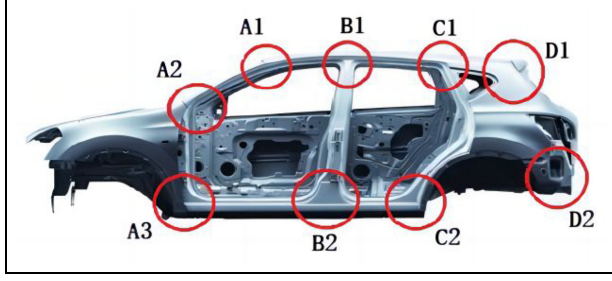


Figure 1. Distribution of joint structure in SUV body.

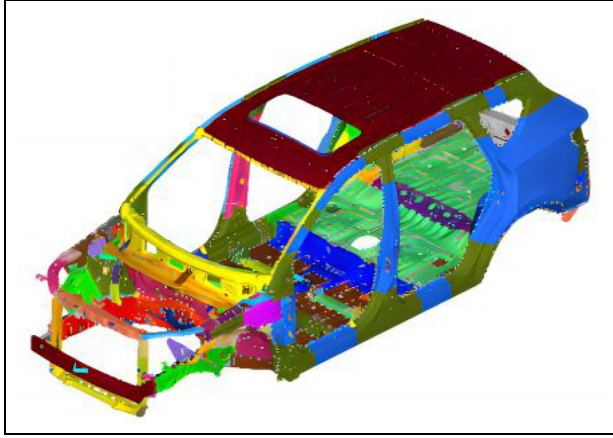


Figure 2. Finite element model of car body.

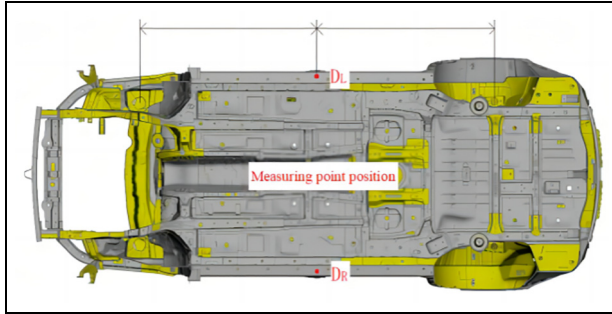


Figure 3. Schematic diagram of deflection measuring point position under bending stiffness analysis.

elastic modulus (E) is $1.87E + 05$ MPa, Poisson's ratio (ν) is 0.3, density (ρ) is $7.80E-09$ g/mm³, and the yield strength is 152 MPa.

Analysis and calculation of body bending stiffness. In this study, the midpoint of the door sill beam, situated at the positions where the front and rear shock absorber mounting points are supported, was chosen as the measurement point. The position of the measurement point is illustrated in Figure 3.

At this juncture, the bending stiffness of the body can be determined by the ratio of the sum of forces (F)

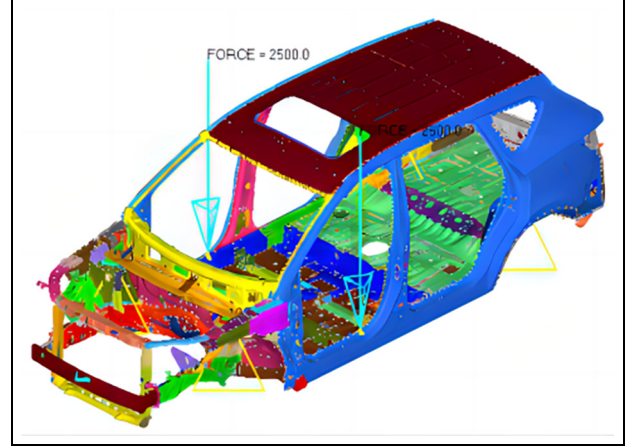


Figure 4. Constraint and loading model under bending stiffness analysis.

to the deflection value at the measurement point on the body's door sill beam. The calculation formula is given as follows:

$$EI = \frac{\sum F}{Z_{middle}} \quad (1)$$

In accordance with the bending stiffness testing methodology employed by a specific company, the body undergoes specified constraints and loading conditions,¹⁶ outlined as follows:

- (1) Constraint conditions: Translational degrees of freedom are constrained in both the left and right front suspensions, as well as the left and right rear suspensions.
- (2) Loading conditions: Equal magnitude loads of 2500 N are applied in the negative z -axis direction, symmetrically positioned at the midpoint of the left and right side door sill beam shock absorber mounting points on the body.

The constrained and loaded body model is illustrated in Figure 4.

The bending deformations of points D_L and D_R are extracted, and their average values are utilized as the vertical displacement of the body. Given deflection values of 0.4547 mm at point D_L and 0.4396 mm at point D_R , the average deflection is calculated to be 0.44715 mm. By substituting the average deflection values of points D_L and D_R into formula (1), we derive:

$$EI = \frac{\sum F}{Z_{middle}} = 11181.39 \text{ N/mm} \quad (2)$$

Therefore, the bending stiffness is calculated to be 11,181.93 N/mm.

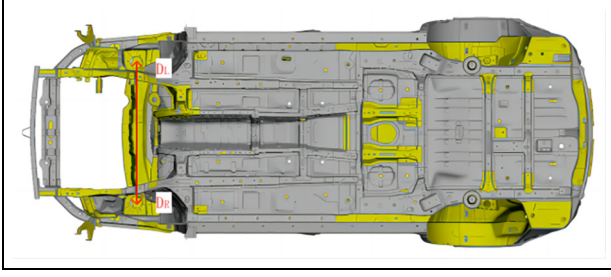


Figure 5. Schematic diagram of deflection measuring point position under torsional stiffness analysis of car body.

Analysis and calculation of body torsional stiffness. During torsional stiffness analysis of the body structure, deflection values can be measured at two specific points, D_L and D_R , situated on the front longitudinal beams. These points are located at the intersection of the two front shock absorbers and the centerline of the front longitudinal beam, as depicted in Figure 5.

The formula for calculating the torsional stiffness of the car body is:

$$GJ = \frac{FL}{\theta} \quad (3)$$

where, GJ represents the torsional stiffness of the car body ($\text{Nm}/^\circ$); F denotes the torsional force applied to the vehicle body (N); L indicates the body wheelbase (mm); and θ represents the relative torsion angle of the loading position ($^\circ$).

Similarly, the body underwent constraint and loading conditions as outlined below:

- (1) Constraint conditions: Constraint the first to third degrees of freedom at the midpoint of the front impact beam. Constraint all translational and rotational degrees of freedom at the mounting holes of the rear shock absorber supports on both the left and right sides.
- (2) Loading conditions: At the center points of the front shock absorber supports on both the left and right sides, Rbe2 elements were established. A pair of equal and oppositely directed forces was applied along the Z-axis direction on each Rbe2 element. To generate a torque of 2000 N·m, considering the 1129.37 mm distance between the two Rbe2 elements at the front shock absorber supports, forces with a magnitude of 1770.9 N were applied.

The constraint and loading model for torsional stiffness analysis is depicted in Figure 6.

Due to the displacement difference between the left and right sides of the body, the torsional angle can be calculated from the displacement difference, as shown in equation (4):

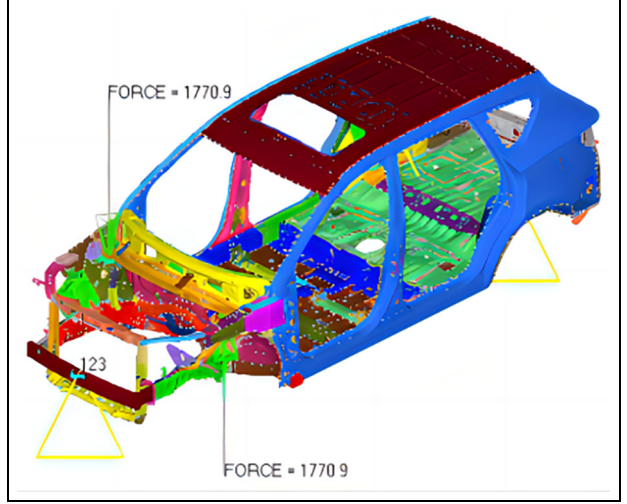


Figure 6. Constraint and loading settings under torsional stiffness analysis.

$$\theta = \arctan \frac{|\partial_A - \partial_B|}{l_1} \quad (4)$$

where θ is represents the torsional angle of the body (rad); ∂_A denotes the absolute value of the Z-axis displacement of left loading point (mm); ∂_B signifies the absolute value of the Z-axis displacement of right loading point (mm); l_1 represents the distance between points A and B (mm).

The measured Z-direction displacements at the loading points on both sides are 1.573 mm and -1.542 mm, respectively, and the calculated relative torsion angle is 0.00276° . By bringing the relative torsion angle into the calculation formula 3, we get:

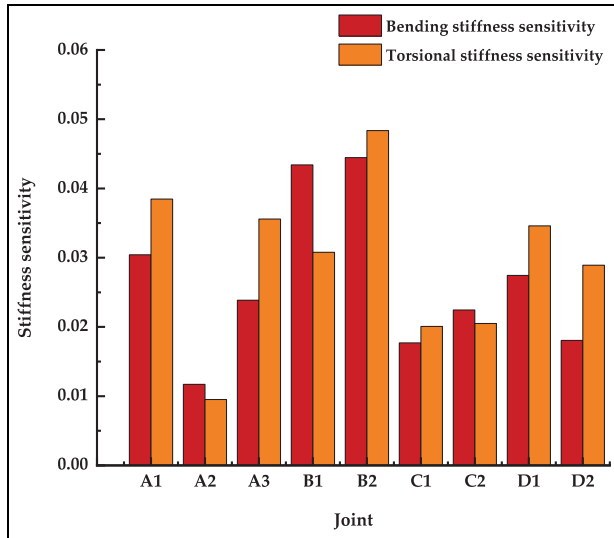
$$GJ = \frac{FL}{\theta} = \frac{M_T}{\theta} = \frac{M_T}{\theta \cdot \frac{180}{\pi}} = 12655.7 \text{ N} \cdot \text{m}/^\circ \quad (5)$$

Therefore, the torsional stiffness is calculated to be $12,655.7 \text{ N} \cdot \text{m}/^\circ$.

Sensitivity analysis involves the examination and analysis of the gradient of the indicators of interest in response to alterations in structural parameters. The magnitude of sensitivity values indicates the influence of each design variable on structural performance, enabling the rapid identification of parameters with significant impact on the system or model. This analysis yields valuable information pertinent to the analytical process.¹⁷ For the purposes of this study, the joint structures on the left side of the body were uniformly selected. The thickness of each joint structure is designated as the design variable. The responses in mass and displacement are defined, with the constraint condition established as the requirement for body bending stiffness or torsional stiffness to exceed the original value. The calculation for sensitivity analysis is conducted

Table 1. Sensitivity value of joint rigidity.

Joint	Bending stiffness sensitivity	Torsional stiffness sensitivity
A1	3.07E-02	3.9E-02
A2	1.2E-02	0.96E-02
A3	2.43E-02	3.6E-02
B1	4.4E-02	3.12E-02
B2	4.5E-02	4.91E-02
C1	1.81E-02	2.05E-02
C2	2.3E-02	2.1E-02
D1	2.8E-02	3.53E-02
D2	1.86E-02	2.95E-02

**Figure 7.** Histogram of joint stiffness sensitivity.

with the aim of minimizing the overall body mass. The data from sensitivity analysis for each joint are acquired through solver calculations and are presented in Table 1. In this context, bending stiffness sensitivity indicates how the body joint structure responds to changes in body bending stiffness, whereas torsional stiffness sensitivity indicates how the body joint structure reacts to variations in body torsional stiffness. For ease of comparison, the sensitivity of each joint structure is graphically represented as a bar chart, as illustrated in Figure 7.

In this study, comprehensive sensitivity was determined by considering both torsional stiffness sensitivity (S1) and bending stiffness sensitivity (S2). The formula for calculating comprehensive sensitivity is as follows:

$$S = 50\% \times S1 + 50\% \times S2 \quad (6)$$

where S represents the comprehensive sensitivity; S1 represents the torsional stiffness sensitivity; S2 represents the bending stiffness sensitivity.

The calculated comprehensive sensitivity priorities for each joint structure are shown in Table 2.

From Table 2, it is evident that the top three priorities in comprehensive sensitivity are B2, B1, and A1. Therefore, A1, B1, and B2 were chosen as the research targets for the next stage.

Simulation analysis of joint deformation in frontal and side collisions

A-pillar joints deform mainly during frontal vehicle collisions, while B-pillar joints deform mainly during side vehicle collisions. Therefore, it is necessary to simultaneously establish frontal and side collision models for the entire vehicle. Frontal and side collision simulations are conducted in accordance with the requirements outlined in China's "C-NCAP Management Regulations (2021 Edition)". In the frontal collision simulation process, a rigid wall, easily constructible due to its simple structure, is selected to replace the rigid barrier. The material of the rigid wall, MAT 2, is 1mm thick and oriented perpendicular to the ground, facing the front of the vehicle. The frontal collision model consists of 1,914,523 nodes and 2,102,552 elements, as shown in Figure 8. The deformable moving barrier comprises a trolley and energy-absorbing blocks. The purpose of the energy-absorbing blocks is to simulate the energy absorption function of the car's bumper. The energy-absorbing block comprises eight elements, divided into two rows, with each row containing four elements. The energy-absorbing block is constructed of honeycomb aluminum material, while the trolley is constructed of rigid body material. The side collision model comprises 2,109,635 nodes and 2,366,524 elements, as illustrated in Figure 9.

A collision speed of 50 km/h and a collision time of 15s were set for the frontal and side collision simulations. In HyperView, kinetic energy, internal energy, system hourglass energy, and system total energy curves during the frontal collision process were extracted, as shown in Figure 10.

From the graph, it can be observed that during the collision of the entire vehicle, the kinetic energy gradually decreases while the internal energy gradually

Table 2. Priorities of different joint structures.

Name of joint structure	A1	A2	A3	B1	B2	C1	C2	D1	D2
Priority	3	9	5	2	1	8	7	4	6

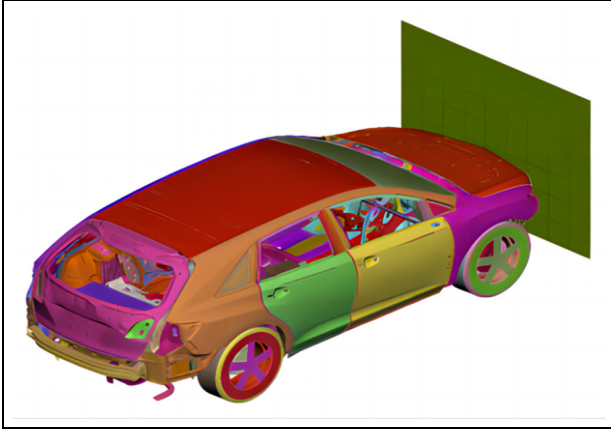


Figure 8. Finite element model of vehicle frontal collision.

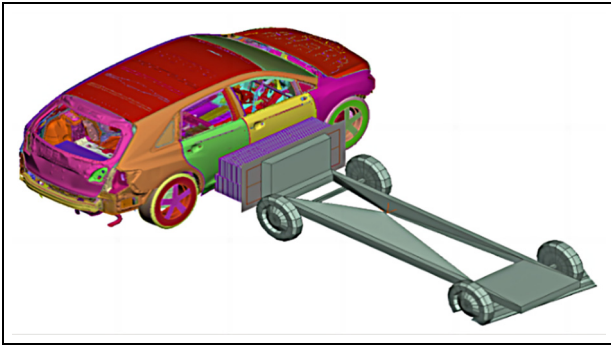


Figure 9. Finite element model of vehicle side impact.

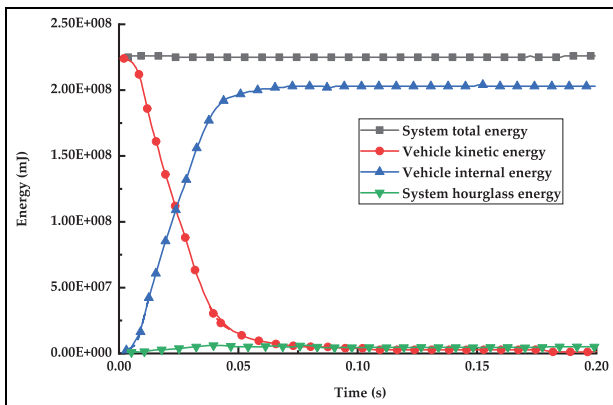


Figure 10. Whole-vehicle frontal collision energy variation curve.

increases. There are no significant jumps in the curves, indicating reasonable changes that align with actual scenarios. Additionally, the ratio of the hourglass energy to the internal energy is 3.8%, meeting the requirement that the hourglass energy should not exceed 5% of the internal energy. Therefore, the total energy remains largely conserved, demonstrating the

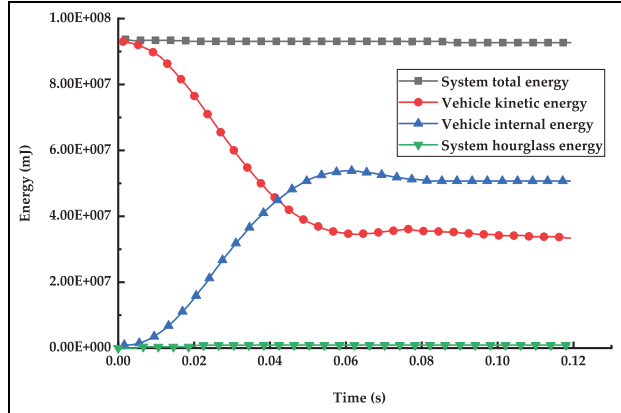


Figure 11. Whole-vehicle lateral collision energy variation curve.

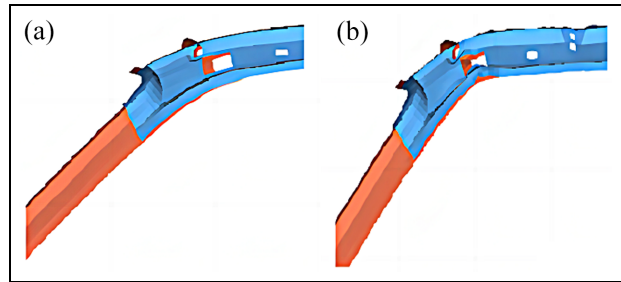


Figure 12. Deformation diagram of joint on A-pillar: (a) before collision and (b) after collision.

reliability and effectiveness of the collision simulation model.

The energy change curve for the entire vehicle during the side collision is shown in Figure 11. The graph illustrates that the System total energy curve remains relatively stable with minimal fluctuations. Furthermore, the kinetic energy curve gradually decreases, and the internal energy curve gradually increases. This is attributed to the MDB model experiencing a decrease in kinetic energy after the collision. Moreover, during the collision process, the vehicle undergoes deformation and absorbs energy, leading to an overall increase in internal energy. The hourglass energy comprises approximately 3.1% of the total energy, falling within an acceptable range, thus aligning with the law of energy conservation. This indicates the credibility of the simulation model.

The joint of the A-pillar experiences deformation both before and after a frontal collision, as shown in Figure 12.

Based on Figure 12, it becomes evident that the deformation of the A-pillar upper joint is not significant. For objectively analyzing the deformation of the A-pillar upper joint, angular measurements can be performed along the A-pillar. Three points, O1, O2, and O3, may be selected on the A and B-pillars.

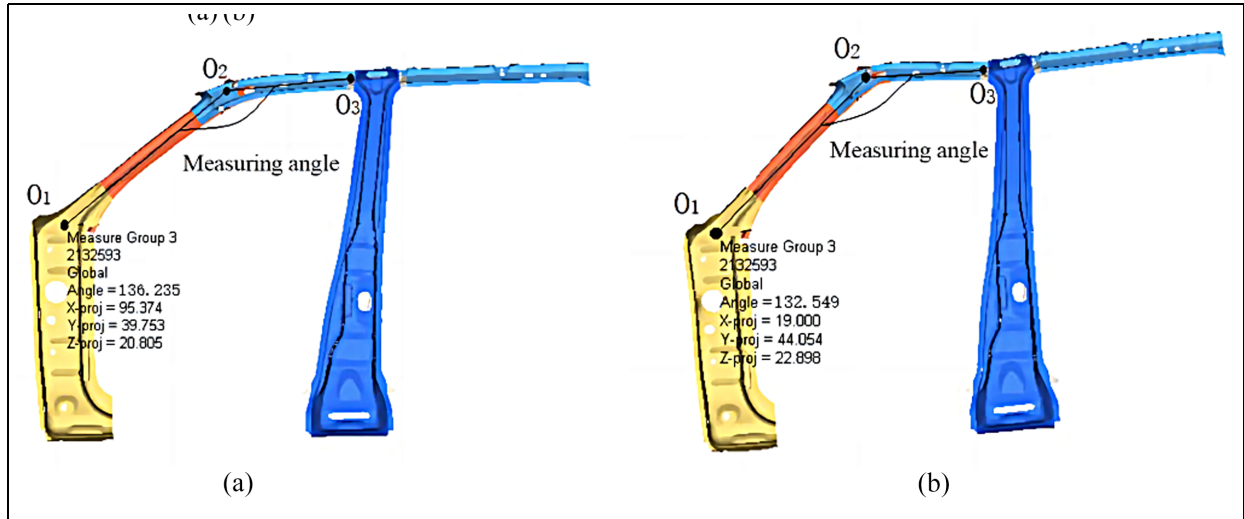


Figure 13. Schematic diagram of measuring angle of A-pillar: (a) before collision and (b) after collision.

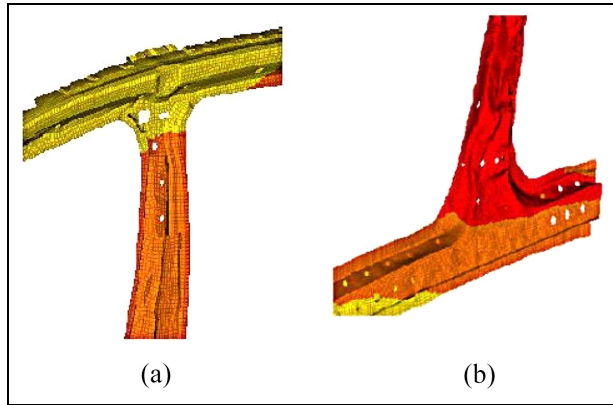


Figure 14. Deformation diagram of upper and lower joints of B-pillar: (a) B-pillar upper joint and (b) B-pillar lower joint.

Subsequently, recording the measurement angles before and after the collision will accurately reflect the bending condition of the A-pillar upper joint. The positions of the three measurement points are illustrated in Figure 13.

Observing Figure 13, it is evident that the angle measurement of the A-pillar was 136.235° before the collision. A smaller measurement angle indicates more severe bending of the A-pillar. Hence, the measurement angle with the smallest value was chosen as the post-collision measurement angle, measuring 132.549° . This corresponded to a decrease of 3.687° . The relatively minor reduction will not cause significant damage to the passenger compartment, thus meeting the requirements.

Deformation diagrams of the upper and lower joints of the B-pillar after the collision are shown in Figure 14.

The deformation of the A-pillar upper joint after the collision has been determined not to cause significant

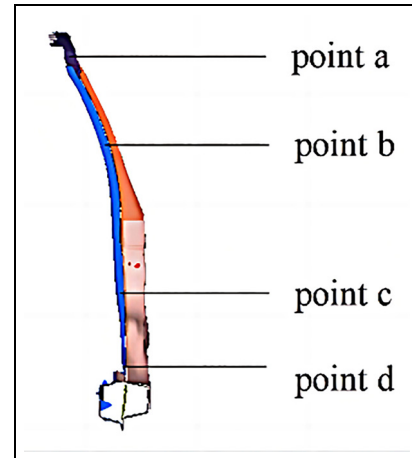


Figure 15. Measuring point of B-pillar.

damage to the passenger compartment according to the collision simulation analysis. The deformation of the B-pillar upper joint area is relatively minor, whereas the deformation in the B-pillar lower joint area is comparatively more substantial. For the purpose of providing a more specific analysis of the deformation of the upper and lower joints of the B-pillar during the collision, the researchers selected four measurement points on the B-pillar from top to bottom and plotted their intrusion curves in the Y-direction (vehicle coordinate system). The measurement points of the B-pillar are shown in Figure 15 and are labeled as points a, b, c, and d. The intrusion curves for each measurement point are shown in Figure 16.

From the figures, it is evident that the maximum deformation occurs at point C, with a peak value of 166 mm, while point D reaches 155 mm, point B attains 133 mm, and point A records the smallest peak value of

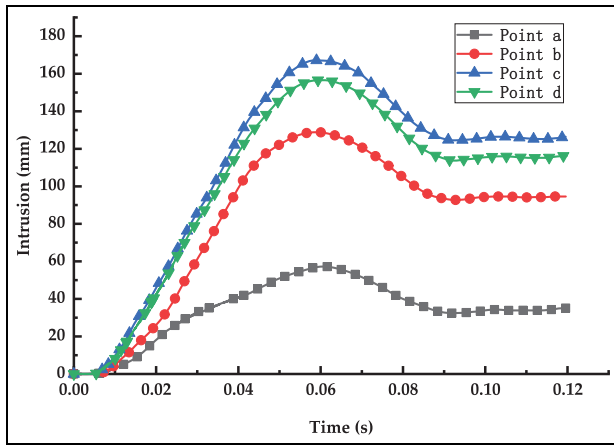


Figure 16. Invasion curve of B-pillar.

Table 3. Relevant parameters of design variables.

Design variable name	Symbolic representation	Initial value (mm)
A-pillar reinforcing plate	t_1	1.2
A-pillar inner plate	t_2	1.2
B-pillar reinforcing plate	t_3	1.5
Door sill beam reinforcement plate	t_4	1.5
B-pillar inner plate	m_1	1
B-pillar reinforcing plate	m_2	1
B-pillar outer plate	m_3	1
Door sill beam reinforcement plate	m_4	3

59 mm. Overall, the area exhibiting the most significant deformation is the lower joint of the B-pillar. The lower joint of the B-pillar is a conventional structural component within the body, and its capacity to withstand lateral loads resulting from side collisions is of paramount safety concern. Excessive deformation in this region poses a potential safety hazard to both drivers and passengers. Hence, it is imperative to mitigate the intrusion of the lower joint of the B-pillar during side collisions, rendering it a primary target for optimization efforts.

Multi-objective optimization design of joints

This paper explores the multi-objective optimization of joint structures with a focus on stiffness and mass as constraints and objectives. The study seeks to optimize these joint structures in terms of materials and thickness to identify the optimal solution.

Multi-objective optimization model

Considering the factors of materials and thickness of sheet metal components at the joint structure locations, eight design variables are defined in this study. This list of design variables includes: A-pillar reinforcement material, B-pillar inner panel material, B-pillar reinforcement material, threshold beam reinforcement material, B-pillar inner panel thickness, B-pillar reinforcement thickness, B-pillar outer panel material, and threshold beam thickness. Considering the range of thickness selection for high-strength steel plates and the plate thickness associated with the vehicle structure, the initial values and value ranges for each design variable are provided in Tables 3 and 4.

Referring to the intrusion response and occupant injury severity in the side collisions of similar vehicle models as per the “C-NCAP Management Rules (2021 Edition)”¹⁸, it is evident that the intrusion at the four points on the B-pillar is significant, presenting a potential risk to the occupants. Therefore, this paper sets the intrusion at these four points as constraint conditions:

$$\begin{cases} 40 \leq L_a \leq 50 \\ 80 \leq L_b \leq 120 \\ 120 \leq L_c \leq 160 \\ 90 \leq L_d \leq 130 \end{cases}$$

The intrusion values at points a, b, c, and d on the B-pillar are denoted as L_a , L_b , L_c , and L_d , respectively.

For the optimization design, the following four indicators are selected to evaluate the lightweighting of the B-pillar lower joint: the total mass (M), the stiffness of the A-pillar upper joint (K_A), the B-pillar upper joint

Table 4. Range of design variables.

t_1 (mm)	t_2 (mm)	t_3 (mm)	t_4 (mm)	m_1	m_2	m_3	m_4
0.7	0.9	1.1	1.3	DC03	DC03	DC03	DC03
0.8	1.0	1.2	1.4	HC260LAD	HC260LAD	HC260LAD	HC260LAD
0.9	1.1	1.3	1.5	HC340LAD	HC340LAD	HC340LAD	HC340LAD
1.0	1.2	1.4	1.6	DP590	DP590	DP590	DP590
1.1	1.3	1.5	1.7	DP780	DP780	DP780	DP780
1.2	1.4	1.6	1.8	BTR165	BTR165	BTR165	BTR165
1.3	1.5	1.7	1.9	HC300D	HC300D	HC300D	HC300D

(K_{B1}), and the B-pillar lower joint (K_{B2}). The optimization objectives can be defined as follows:

$$\begin{cases} \min mass \\ \max K_A \\ \max K_{B1} \\ \max K_{B2} \end{cases}$$

In multi-objective optimization scenarios, the weighting method can improve solution efficiency. It considers the balance among multiple objectives from a global perspective, thus preventing the occurrence of local optima.¹⁹ The mathematical expressions are delineated below:

$$\max F(x) = \omega_1 f_1(x) + \omega_2 f_2(x) + \omega_3 f_3(x) + \omega_4 f_4(x) \quad (7)$$

where, $F(x)$ is a single-objective optimization function, while $f_1(x) = -M$, $f_2(x) = K_A(x)$, $f_3(x) = K_{B1}(x)$, and $f_4(x) = K_{B2}(x)$; denote the objective functions. The symbol ω_i denotes the weight coefficient, confined within the interval of $0 \leq \omega_i \leq 1$, subject to the constraint of $\omega_1 + \omega_2 + \omega_3 + \omega_4 = 1$. The weight coefficients selected for this study are $\omega_1 = \omega_2 = \omega_3 = \omega_4 = 0.25$.

Multi-objective optimization model solution

Orthogonal experimental design. Orthogonal experimental design is a method that utilizes orthogonal tables to select experimental schemes characterized by “uniform distribution” and “strong representativeness.” It allows for the simultaneous study of multiple levels and factors and is known for its high reliability and cost-effectiveness in optimization design.²⁰ This paper defines eight variables, namely t_1 – t_4 and m_1 – m_4 , each comprising seven levels, yielding 49 sets of experimental samples. Based on the data from each set of experimental samples, the attributes of the design variables are sequentially modified in ANSA. Subsequently, the modified problem is submitted to Optistruct for computational solution, necessitating 49 calculations in total.

Response surface (RSM) approximation model. The response surface method (RSM) is utilized to fit complex relationships and has demonstrated high effectiveness.²¹ The precision of the response surface model is evaluated using two parameters: the coefficient of determination (R^2) and the adjusted coefficient of determination (R_{adj}^2). If R^2 and R_{adj}^2 are closer to 1, this indicates that the RSM model accurately predicts responses at the sample points, closely approximating the true values. The formulas for calculating R^2 and R_{adj}^2 are specified in equations (8) and (9), respectively:

$$R^2 = \frac{\sum_{i=1}^p (\tilde{y}_i - \bar{y}_i)^2}{\sum_{i=1}^p (y_i - \bar{y}_i)^2} \quad (8)$$

$$R_{adj}^2 = 1 - \frac{\sum_{i=1}^p (\tilde{y}_i - \bar{y}_i)^2 (p-1)}{\sum_{i=1}^p (y_i - \bar{y}_i)^2 (p-k-1)} \quad (9)$$

where y_i represents the true values at the sample points; $\tilde{y}_i$ represents the predicted response values obtained from the response surface; $\bar{y}_i$ represents the average of the true values at the sample points; k represents the degrees of freedom; p represents the number of collected sample points.

In this paper, our approximation model incorporates a total of eight variables, with output responses comprising joint mass and the torsional stiffness of the A-pillar, the B-pillar upper joint, and the B-pillar lower joint. The variables and responses are represented as follows: x_i ($i = 1, 2, \dots, 8$), y_1 (joint mass), y_2 (A-pillar torsional stiffness), y_3 (B-pillar upper joint torsional stiffness), and y_4 (B-pillar lower joint torsional stiffness). Based on vehicle frontal and side collision simulation conditions, the polynomial expressions for the responses are as follows:

$$y_1 = 6.356 + 1.526x_1 + 1.377x_2 + 2.117x_3 + 0.024x_4 \\ + 0.043x_5 + 0.029x_6 + 0.053x_7 + 0.088x_8$$

The determination coefficient R^2 and adjustment coefficient R_{adj}^2 of y_1 are respectively: $R^2 = 0.9748$, $R_{adj}^2 = 0.9627$

$$y_2 = -71 + 131x_1 + 89.6x_2 - 23.9x_1x_2 - 141.9x_1x_3 \\ + 41.83x_1x_4 - 385.85x_1^2 - 45.8x_7^2 - 43.3x_8^2$$

The determination coefficient R^2 and adjustment coefficient R_{adj}^2 of y_2 are respectively: $R^2 = 0.9971$, $R_{adj}^2 = 0.9812$

$$y_3 = -69 + 76.9x_1 + 52.3x_2 - 54x_1x_2 - 25.8x_1x_3 \\ + 84.6x_1x_4 - 241.63x_1^2 - 49.7x_7^2 - 65.2x_8^2$$

The determination coefficient R^2 and adjustment coefficient R_{adj}^2 of y_3 are respectively: $R^2 = 0.9829$, $R_{adj}^2 = 0.9714$

$$y_4 = -16 + 211.2x_1 + 74.5x_2 - 23.9x_1x_2 - 42.3x_1x_3 \\ + 98.53x_1x_4 - 257.1x_1^2 - 89.2x_7^2 - 55.7x_8^2$$

The determination coefficient R^2 and adjustment coefficient R_{adj}^2 of y_4 are respectively: $R^2 = 0.9815$, $R_{adj}^2 = 0.9733$

In this section, only a subset of terms in the response polynomials for y_1 , y_2 , y_3 , and y_4 are presented. This is

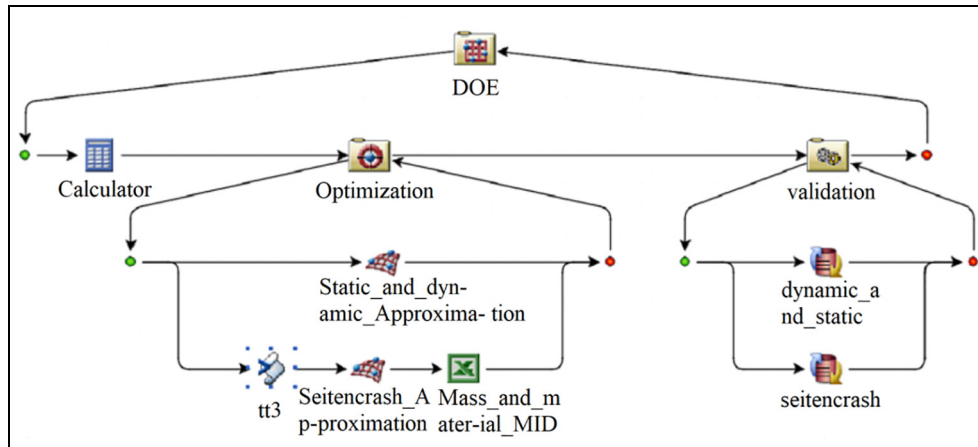


Figure 17. The optimization process in Isight.

because each approximation model involves polynomial terms of eight component variables. Listing the second-order polynomial terms for all eight variables would occupy excessive space. Therefore, only a few terms from the front and rear of the polynomials are included.

From the fitting results, it is evident that both the coefficient of determination R^2 and the adjusted coefficient of determination R_{adj}^2 are close to 1. This indicates that the response surface models exhibit a high level of fit accuracy at the sampled points and can be used as substitutes for complex real-world models.

Solution of the multi-objective optimization model. Building upon the polynomial representations for y_1 , y_2 , y_3 , and y_4 , this study employs a simulated annealing algorithm in the Isight software for multi-objective optimization design. In the optimization problem, the simulated annealing algorithm is characterized by lowering the system temperature until it converges to a feasible and stable solution. Throughout this process, at higher temperatures, suboptimal solutions relative to the current one are accepted to ensure the algorithm escapes local optima. Simultaneously, by exploring the solution space and gradually reducing the probability of accepting inferior solutions, the algorithm converges gradually to a region for finding the optimal solution.²² The optimization flowchart of Isight is shown in Figure 17.

The essential parameter configurations for the simulated annealing optimization algorithm are depicted in Table 5.

The thickness and material of each variable before and after optimization are illustrated in Table 6.

Based on the data in Table 6, a reduction in the thickness of components, accompanied by corresponding changes in materials, is evident. For components with a significant impact on side collisions, such as the

B-pillar reinforcement panel and B-pillar inner panel, a decrease in thickness is correlated with an upgrade in material grade, transitioning from conventional high-strength steel to ultra-high-strength steel.

Analysis of optimization results

Based on the parameters of the optimization plan, this study made material and thickness adjustments to the vehicle's finite element model while keeping other attributes unchanged. Subsequently, in accordance with the stiffness simulation constraints and loading requirements mentioned earlier, the stiffness of the optimized vehicle was recalculated to verify whether the vehicle's performance has genuinely improved. Table 7 presents a comparison of the vehicle's stiffness before and after optimization.

Table 7 illustrates that the bending stiffness of the vehicle body post-optimization is 12,223.1 N/mm, reflecting a 9.31% improvement over the pre-optimization state. Moreover, the torsional stiffness post-optimization measures 14,094.5, denoting an 11.37% increase compared to the pre-optimization condition. These findings underscore the substantial enhancement of the vehicle body's stiffness through the optimization approach and confirm the pivotal role of joints in influencing the vehicle body's stiffness.

Prior to optimization, the entire vehicle model manifested issues, including severe deformation of the B-pillar lower joint and significant intrusion of the B-pillar. Subsequently, adjustments were made to the side-impact finite element model, and additional side-impact simulations were performed to further evaluate the impact of the optimization approach on side-impact collisions.

Figure 18 depicts the B-pillar deformation results before and after optimization. The figure clearly indicates that post-optimization, the B-pillar deformation

Table 5. Configuration of main parameters of simulated annealing algorithm.

Parameter name	Value	Parameter description
Maximum number of iterations	5000	Estimated maximum number of iterations
Convergence criterion	$\leq 1.0E-8$	Is the difference between the feasible solution and the optimal solution convergent
Convergence check interval	5	If the adjacent continuous residuals are smaller than the parameters of the convergence criterion, Optimization stops
Maximum number of allowed calculation failures	4	Exceeding this criterion indicates a failed calculation

Table 6. Changes of plate thickness and material before and after optimization.

Part name	Initial value	Optimized value
A-pillar stiffener (mm)	1.2	1.0
B-pillar inner plate (mm)	1.2	1.1
B-pillar stiffener (mm)	1.5	1.3
Sill beam reinforcing plate (mm)	1.5	1.3
B-pillar inner plate	DC03	DP590
B-pillar stiffener	DC03	DP780
B-pillar outer plate	DC03	BTR165
Sill beam reinforcing plate	HC340LAD	HC300D

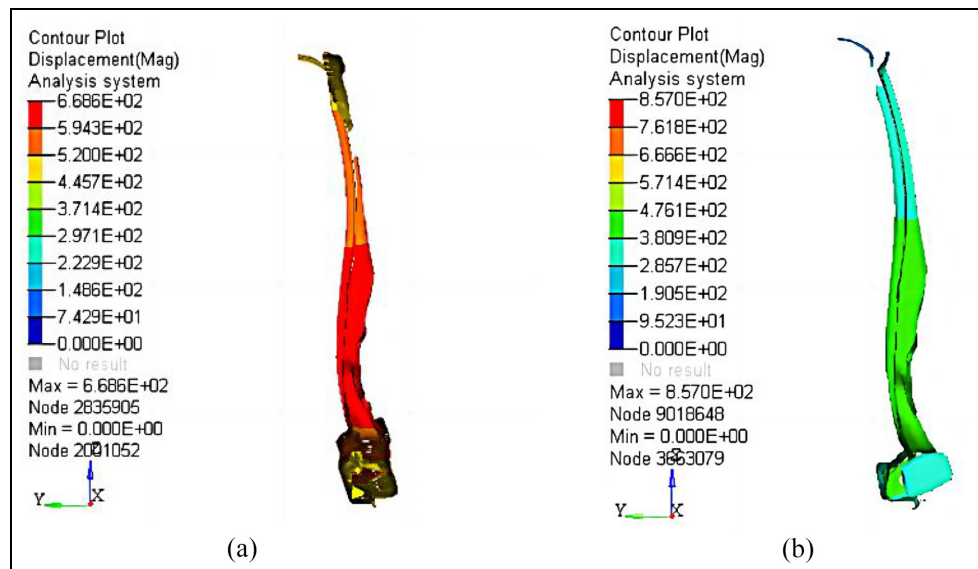
Table 7. Comparative analysis of bending stiffness and torsion stiffness of car body before and after optimization.

	Value before optimization	Optimized value	Ascending proportion (%)
Bending rigidity N/mm	11,181.9	12,223.1	9.31
Torsional rigidity N m/°	12,655.7	14,094.5	11.37

is significantly diminished, suggesting that this optimization approach effectively mitigates B-pillar deformation in side-impact collisions.

To further analyze the intrusion at the B-pillar points a, b, c, and d after optimization, measurements were taken at these four points before optimization to assess intrusion levels. Figure 19 presents a comparative illustration of the intrusion levels at B-pillar points a, b, c, and d before and after optimization.

As shown in Figure 19, the intrusion levels at B-pillar points a, b, c, and d are reduced after optimization. Specifically, point a decreased from 59 mm to 46 mm, a reduction of 13 mm; point b decreased from 132 mm to 76 mm, a reduction of 56 mm; point c decreased from 166 mm to 137 mm, a reduction of 29 mm; and point d decreased from 155 mm to 133 mm, a reduction of 22 mm. Before optimization, there was significant intrusion at the B-pillar lower joint, primarily at points c and d. After optimization, both points c and d saw substantial reductions in intrusion, improving the overall B-pillar structure's intrusion levels in side-impact collisions and effectively enhancing the vehicle's crashworthiness.

**Figure 18.** Deformation plot of B-pillar before and after optimization: (a) before optimization and (b) after optimization.

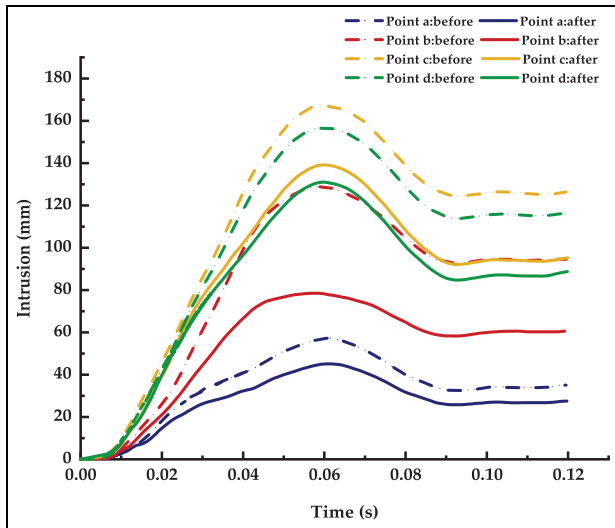


Figure 19. Comparative analysis of B-pillar invasion before and after optimization.

Taking into account the simulation results for various performance metrics, it is evident that the vehicle's stiffness and crashworthiness are improved after optimization. This indicates that the joint optimization approach is reasonable and viable.

Discussion

In this study, it was found that the lower joint of the B-pillar plays a crucial role in enhancing the rigidity and crashworthiness of the vehicle. Firstly, within the vehicle's side structure, the B-pillar and its lower joint play a significant load-bearing role. The stiffness of the lower joint significantly impacts the overall stiffness of the vehicle. Secondly, in frontal and side collisions, the vehicle's side structure has less energy absorption structure and space compared to the front structure. The lower joint of the B-pillar serves as a primary load-bearing component in side impacts, and any deformation thereof is more likely to endanger occupants. Therefore, optimizing the lower B-pillar joint can effectively enhance the rigidity and crashworthiness of the vehicle, aligning with prior research on vehicle joints.¹⁸

An SUV body comprises a total of 18 joint structures on each side. Selecting joint structures for optimization solely based on empirical methods can be resource-intensive, potentially impacting project timelines and complicating the optimization process. Therefore, this study involved conducting a sensitivity analysis on the 18 joints, focusing on the vehicle's bending and torsional stiffness, facilitating the selection of key joint structures highly sensitive to these factors. Additionally, using collision simulations, we pinpointed the lower B-pillar joint as experiencing the most

significant deformation in collisions and subsequently optimized it, achieving maximal optimization with minimal cost.

This study optimized the material and thickness of the lower joint of the B-pillar through systematic finite element analysis and multi-objective optimization methods, thereby enhancing the stiffness and crashworthiness of the vehicle body structure. Our research provides quantitative data support and specific optimization solutions, making the optimization results more reliable and practically guided. Our work not only improves the performance of the vehicle body structure but also provides valuable experience and methods for future similar studies.

This study has limitations. In the preliminary stage of the research, we did not investigate the impact of multi-material composite joints on the vehicle's crashworthiness. Today, the trend of optimizing vehicle structures by combining traditional metals with polymer composite materials for strength, weight, and durability is becoming increasingly evident. Studying how to use multi-material composite joints to achieve vehicle lightweighting while improving crashworthiness is essential. Further research is needed to assess the impact of multi-material composite lower B-pillar joints on the vehicle's crashworthiness.

In summary, this study significantly enhances the crashworthiness of the vehicle. In future research, integrating the investigation of multi-material composite joints' impact on vehicle crashworthiness with research on vehicle lightweighting is of considerable academic and practical importance.

Conclusion

This study conducted sensitivity analysis and collision simulation to identify the critical joint in the vehicle body, namely the lower joint of the B-pillar. Subsequently, multi-objective optimization was applied to enhance the stiffness and crashworthiness effectively, thereby providing a reference for automotive design. The specific conclusions are as follows:

- (1) Utilizing finite element theory and analysis procedures, preprocessing was conducted on the vehicle's structure. Subsequently, the bending stiffness measured 11,181.93 N/mm, while the torsional stiffness was determined to be 12,655.7 N m/°.
- (2) Sensitivity analysis was performed using joint component plate thickness as design parameters to identify critical joints impacting the vehicle's bending and torsional stiffness, such as the upper A-pillar joint, upper B-pillar joint, and lower B-pillar joint, thus offering valuable insights for subsequent optimization efforts.

- (3) The analysis of simulation results from frontal and side-impact collision models served to validate the efficacy of the finite element models. Upon analyzing the collision results, it was concluded that the upper A-pillar joint experienced minimal bending deformation, while the upper B-pillar joint exhibited relatively minor deformation, reaching a maximum intrusion of 133 mm. In contrast, the lower B-pillar joint demonstrated more significant deformation, with a maximum intrusion of 166 mm.

Declaration of conflicting interests

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References

1. Ki WY, Heo SJ, Kang DO, et al. Development of knowledge based body structure concept design model. *Int J Automot Tech* 2017; 18: 429–437.
2. Zhang Y and Cao Y. Modal optimization of BIW based on concept feature model. *J Chongqing Jiaotong Univ* 2019; 38(07): 114–118.
3. Izanloo M and Khalkhali A. Optimal design of automotive body B-pillar using simplified finite element model of body-in-prime combined with an optimization procedure. *J Cent South Univ* 2022; 29(12): 3939–3955.
4. Yin C, Lv Z, Wang Y, et al. Mathematical problems in engineering joint structure stiffness analysis of an automotive body in white with the finite element method. *Math Probl Eng* 2022; 2022: 1–11.
5. Hou W, Xu X, Han X, et al. Multi-objective and multi-constraint design optimization for hat-shaped composite T-joints in automobiles. *Thin-Walled Struct* 2019; 143: 106232.
6. Jia L, Song K, Cheng A, et al. Fatigue analysis of auto body T-joints based on virtual test field. *Automot Eng* 2011; 33(5): 423–427.
7. Kiani M, Motoyama K, Rais-Rohani M, et al. Joint stiffness analysis and optimization as a mechanism for improving the structural design and performance of a vehicle. *Proc IMechE Part D: J Automobile Engineering* 2014; 228(6): 689–700.
8. Gaeta A, La Monaca A and Frascá F. A combined optimisation of automotive body joints. SAE technical paper 2003-01-2797, 2003.
9. Tun B and Endur P. A new methodology to determine the design sensitivity of critical automotive body joints for basic design cycle. *Proc IMechE Part D: J Automobile Engineering* 2018; 233(10): 2559–2571.
10. Xiong F, Wang D, Chen S, et al. Multi-objective lightweight and crashworthiness optimization for the side structure of an automobile body. *Struct Multidiscip Optim* 2018; 58: 1823–1843.
11. Pyrz M, Krzywobłocki M and Wolska N. Optimal crashworthiness design of vehicle S-frame using macro-element method and evolutionary algorithm. *Struct Multidiscip Optim* 2022; 65(3): 80.
12. Kongwat S, Jongpradist P and Hasegawa H. Lightweight bus body design and optimization for rollover crashworthiness. *Int J Automot Tech* 2020; 21: 981–991.
13. Zhang D, Mo Y, Ding M, et al. Optimization of rollover crashworthiness and vehicle mass based on unreplicated saturated factorial design. *Appl Sci* 2023; 13(12): 6963.
14. Wang D, Li S and Xie C. Crashworthiness optimisation and lightweight for front-end safety parts of automobile body using a hybrid optimisation method. *Int J Crashworthiness* 2022; 27(4): 1193–1204.
15. Zou S, Dai S, Zhao W, et al. Multi-objective crashworthiness optimization of vehicle body with negative Poisson's ratio structure based on factorial analysis. *Proc IMechE Part D: J Automobile Engineering* 2020; 234(14): 3329–3346.
16. He N and Chen R. Analysis of influencing factors of body-in-white stiffness of commercial vehicle. *Automobile Appl Technol* 2019; (12): 88–90.
17. Loubière P, Jourdan A, Siarry P, et al. A sensitivity analysis method for driving the Artificial Bee Colony algorithm's search process. *Appl Soft Comput* 2016; 40: 515–531.
18. Zhou Z, Deng Y and Peng Y. Optimization of an automobile's safety performance for side impact. *Pop Sci Technol* 2019; 21(3): 33–36.
19. Mi C, Liu J, Zhang C, et al. Reliability analysis and optimization design of magnetic fluid dynamic seal shell structure under thermal/mechanical load. *J Magn Magn Mater* 2024; 594: 172027.
20. Yan Q, Gao ZH and Song ZC. Sensitivity analysis of concrete face rock-fill dam parametric based on the intelligent algorithm. *Appl Mech Mater* 2014; 580–583: 1940–1945.
21. Zhu B, Wang X, Tan L, et al. Optimization design of a reversible pump-turbine runner with high efficiency and stability. *Renew Energy* 2015; 81: 366–376.
22. Kirkpatrick S. Optimization by simulated annealing: quantitative studies. *J Stat Phys* 1984; 34(5): 975–986.